

Vitamin D Deficiency and Rickets

Elevated serum alkaline phosphatase (AP) levels and low serum 25-hydroxyvitamin D levels are often useful laboratory indicators of vitamin D deficiency. In the early stages of rickets, parathyroid hormone (PTH) levels increase to compensate for declining serum calcium levels. However, this compensatory mechanism becomes inadequate if the deficiency persists.

Treatment

The recommended daily intake of vitamin D is 5 to 10 μg . Treatment includes oral vitamin D repletion, typically with cholecalciferol or ergocalciferol (not dihydroxyvitamin D, which is used in specific metabolic disorders), along with a calcium-rich diet. Supplementation with 200 to 400 μg of vitamin D per day until resolution of symptoms—usually within 2 to 3 months—is generally adequate. Additional therapy may include judicious sun exposure to promote endogenous vitamin D synthesis.

Vitamin E (Tocopherol)

Vitamin E deficiency or excess is rarely encountered in clinical practice. Vitamin E is found in vegetable oils, shortenings, fortified grains, dark green leafy vegetables, legumes, nuts, avocados, and small oily fish such as herring and sardines.

As a fat-soluble vitamin, excessive intake of vitamin E can enhance the effects of anticoagulant medications, increasing the risk of purpura and hemorrhage.

Deficiency states are uncommon but can occur. **Ataxia with isolated vitamin E deficiency** (AVED) is a rare, autosomal recessive spinocerebellar neurodegenerative disorder caused by mutations in the *α -tocopherol transfer protein* gene. This defect impairs the transfer of α -tocopherol from lysosomes to lipoproteins, leading to impaired vitamin E transport in the body. The resulting

deficiency predisposes cells to oxidative stress, particularly affecting neurons, and leads to progressive neurologic degeneration if untreated.

Vitamin K (Phytonadione)

Etiology and Pathogenesis

Vitamin K is an essential cofactor for the carboxylation of glutamate residues on several coagulation factors, including factors II (prothrombin), VII, IX, and X, as well as proteins C and S—proteins critical for normal blood clotting.

Dietary vitamin K (phylloquinone) is primarily obtained from green leafy vegetables, certain legumes, soybeans, cereals, and beef liver. Phylloquinone is actively absorbed in the distal small intestine. Approximately half of the body's vitamin K requirement is met through diet, while the other half is synthesized by gut microbiota in the colon.